

Welcome to CSC331 Fall 2025 Data Structures

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Borough of Manhattan Community College/CUNY

CSC331 Spring 2025

Data Structures

Week 7:

Graph definitions
Prim's MST algorithm

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Reminders

- No class Mon 10-20-2025
- Follow Monday schedule on: Fri 10-24-2025
- Program 3 due date
 - 1159pm Fri 10-31-2025
 - late submission receives no credit
- Exam 2
 - Section 501 Tue 11-11-2025 – Zoom 630pm-730pm
 - Section 172 Wed 11-12-2025 – In person room M-1001 630pm-730pm
 - Closed book, closed notes, no devices
 - Section 500 Wed 11-12-2025 – In person room N-453 1010am-1110am
 - Closed book, closed notes, no devices
- Program 4 due date
 - 1159pm Fri 11-21-2025
 - late submission receives no credit
- Posted on Brightspace
 - Week 7 lecture slides

Graphs

- Sample graphs
 - $A=(V_a, E_a)$:
 - $V_a=\{1,2,3,4,5\}$
 - $E_a=\{(1,2),(1,3),(1,4),(2,5),(3,4),(3,5)\}$
 - $B=(V_b, E_b)$:
 - $V_b=\{1,2,3,4,5,6\}$
 - $E_b=\{(1,2),(1,3),(2,4),(2,5),(3,6)\}$
 - $C=(V_c, E_c)$:
 - $V_c=\{\text{Chicago, LA, Miami, NY, Omaha}\}$
 - $E_c=\{(\text{Chicago, LA}), (\text{Chicago, Miami}), (\text{Chicago, NY}), (\text{Chicago, Omaha}), (\text{LA, NY}), (\text{LA, Omaha}), (\text{Miami, NY})\}$

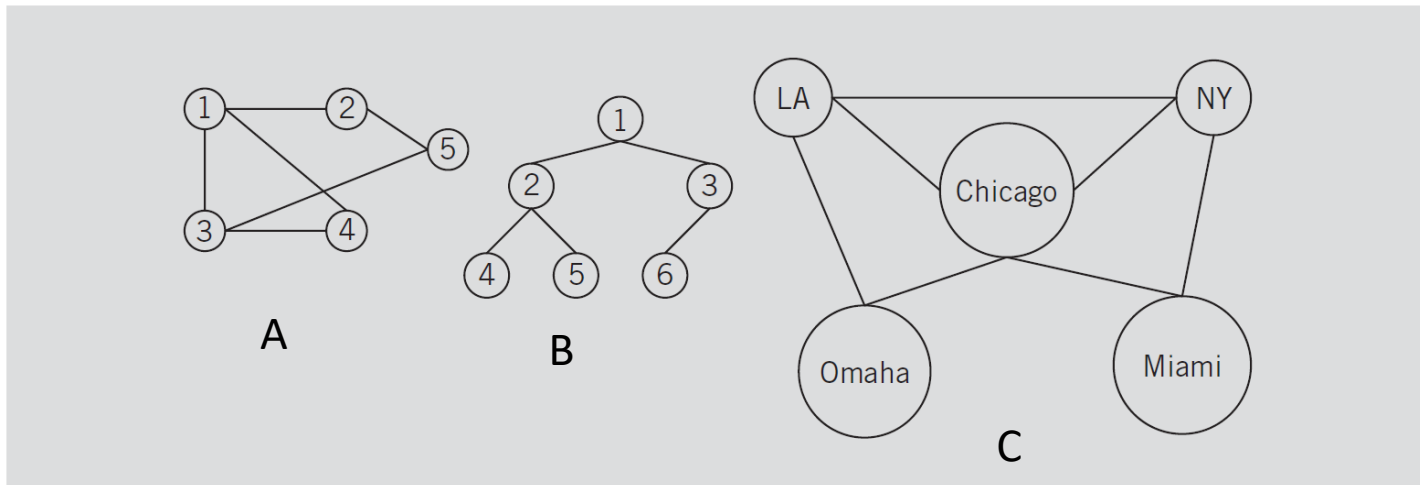


FIGURE 12-3 Various undirected graphs

Graphs

- Sample graphs

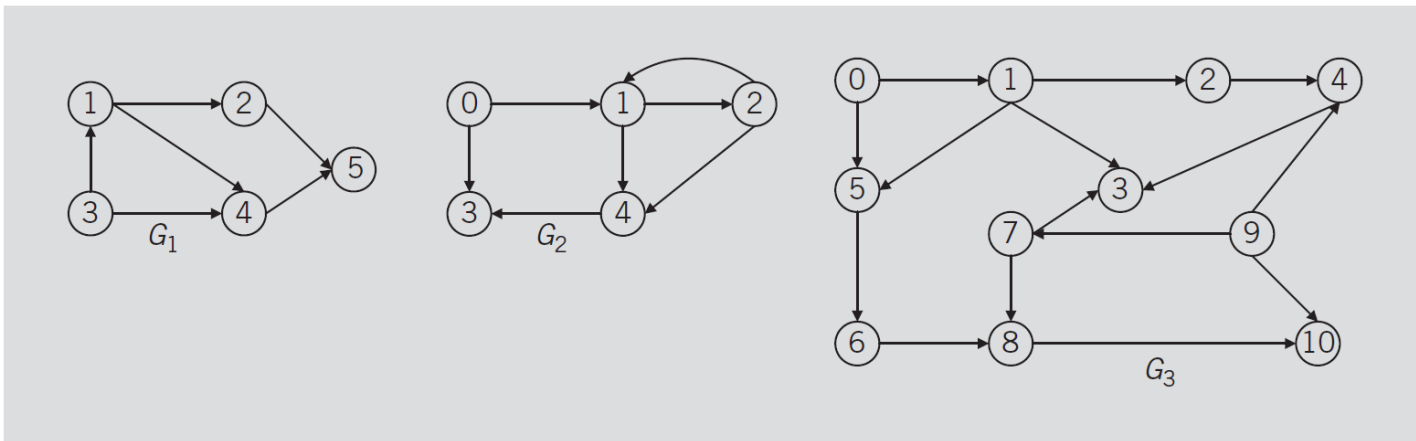


FIGURE 12-4 Various directed graphs

For the graphs of Figure 12-4, we have

$$\begin{aligned}
 V(G_1) &= \{1, 2, 3, 4, 5\} & E(G_1) &= \{(1, 2), (1, 4), (2, 5), (3, 1), (3, 4), (4, 5)\} \\
 V(G_2) &= \{0, 1, 2, 3, 4\} & E(G_2) &= \{(0, 1), (0, 3), (1, 2), (1, 4), (2, 1), (2, 4), (4, 3)\} \\
 V(G_3) &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} & E(G_3) &= \{(0, 1), (0, 5), (1, 2), (1, 3), (1, 5), (2, 4), (4, 3), \\
 & & & (5, 6), (6, 8), (7, 3), (7, 8), (8, 10), (9, 4), \\
 & & & (9, 7), (9, 10)\}
 \end{aligned}$$

Graphs

- Graph representation: adjacency matrix

Consider the directed graphs of Figure 12-4. The adjacency matrices of the directed graphs G_1 and G_2 are as follows:

$$A_{G_1} = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad A_{G_2} = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Graphs

Edges in an undirected graph where $V=\{a,c,e,g,i\}$
(edges in red are duplicate edges)

	a	c	e	g	i
a	(a,a)	(a,c)	(a,e)	(a,g)	(a,i)
c	(c,a)	(c,c)	(c,e)	(c,g)	(c,i)
e	(e,a)	(e,c)	(e,e)	(e,g)	(e,i)
g	(g,a)	(g,c)	(g,e)	(g,g)	(g,i)
i	(i,a)	(i,c)	(i,e)	(i,g)	(i,i)

$$\begin{aligned}\text{Number of unique edges} &= |V|^2 - (|V|(|V|-1))/2 \\ &= 5^2 - (5(5-1))/2 \\ &= 15\end{aligned}$$

Graphs

- Graph representation: adjacency list

Consider the directed graphs of Figure 12-4. Figure 12-5 shows the adjacency list of the directed graph G_2 .

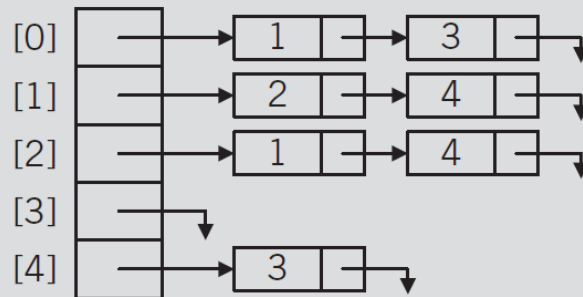


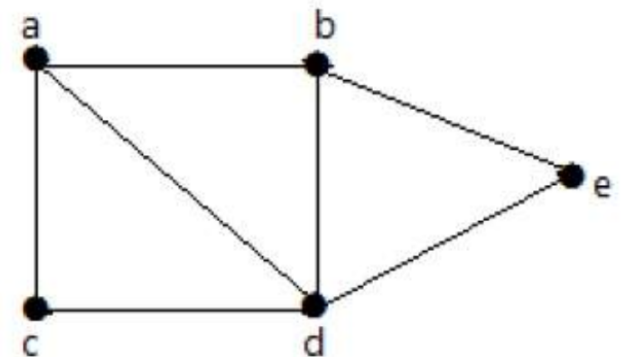
FIGURE 12-5 Adjacency list of graph G_2 of Figure 12-4

Graphs

- A graph is said to be connected if there is a path (i.e., set of edges) between every pair of vertices
- In a connected graph, from every vertex to any other vertex, there should be some path to traverse

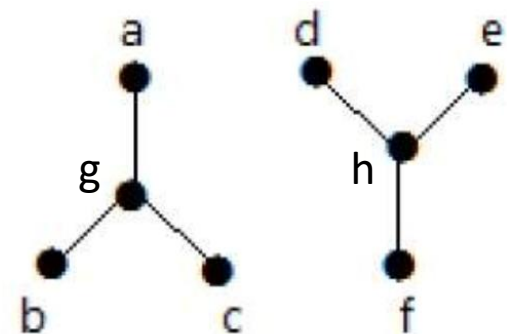
- Sample connected graph, $J=(V_j, E_j)$:

- $V_j=\{a,b,c,d,e\}$
- $E_j=\{(a,b),(a,c),(a,d),(b,d),(b,e),(c,d),(d,e)\}$



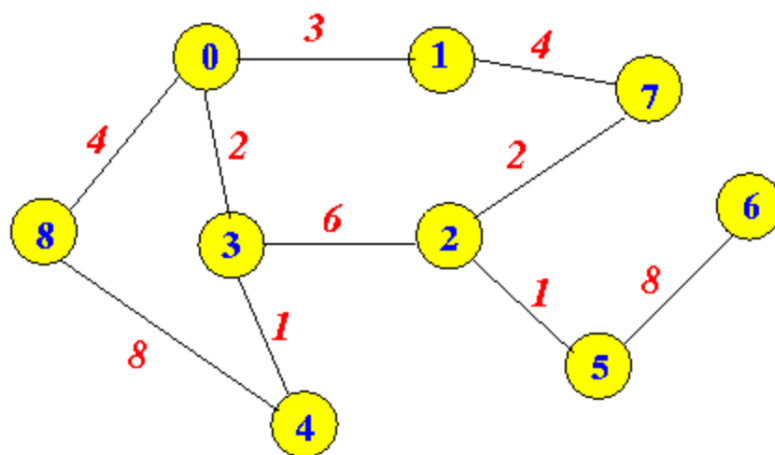
- Sample unconnected graph, $K=(V_k, E_k)$:

- $V_k=\{a,b,c,d,e,f,g,h\}$
- $E_k=\{(a,g),(b,g),(c,g),(d,h),(e,h),(f,h)\}$



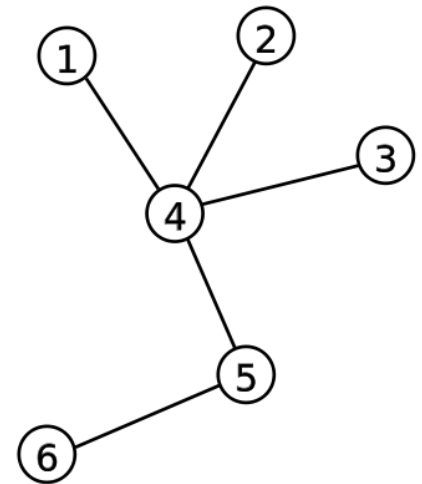
Graphs

- A graph is said to be weighted if each edge of has an associated numerical value, called a weight
- A weight may represent a value such as cost, distance, etc.
- A weighted graph may be directed or undirected
- Sample weighted graph, $G=(V_g, E_g)$:
 - $V_g=\{0,1,2,3,4,5,6,7,8\}$
 - $E_g=\{(0,1,3),(0,3,2),(0,8,4),(1,7,4),(2,3,6),(2,5,1),(3,4,1),(4,8,8),(5,6,8)\}$



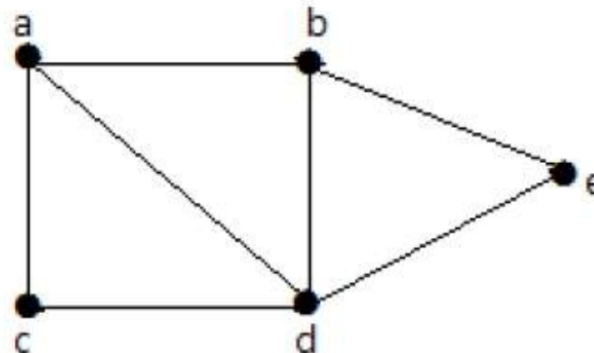
Graphs

- A graph is said to be a tree if any two vertices are connected by exactly one path
- Every acyclic connected graph is a tree, and vice versa.
- A tree with n vertices has $(n-1)$ edges
- Sample tree, $T=(V_t, E_t)$:
 - $V_t=\{1,2,3,4,5,6\}$
 - $E_t=\{(1,4),(2,4),(3,4),(4,5),(5,6)\}$
- In a tree, one vertex may be named the root



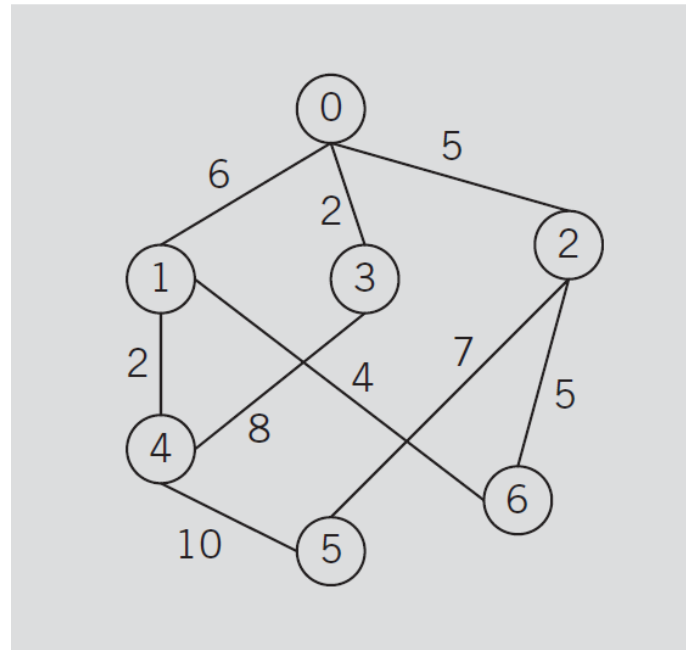
Graphs

- A subgraph of a graph G is another graph, H , formed from a subset of the vertices and edges of G
- The vertex subset must include all endpoints of the edge subset
- A spanning subgraph is a subgraph that contains all the vertices of the original graph
- A spanning tree is a spanning subgraph that is also a tree
- Find a spanning subgraph, $Q_s = \{V_s, E_s\}$, and a spanning tree, $Q_t = (V_t, E_t)$, of this graph: $Q = \{V_q, E_q\}$



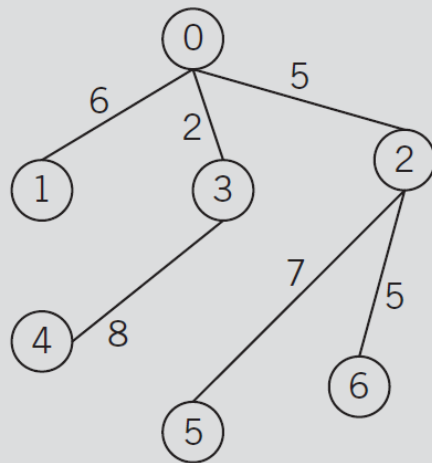
Graphs

- A minimum spanning tree (MST) is a spanning tree of a connected, weighted graph and which has the minimum total edge weight among all spanning trees of the original graph
- Find the minimum spanning tree of this graph:

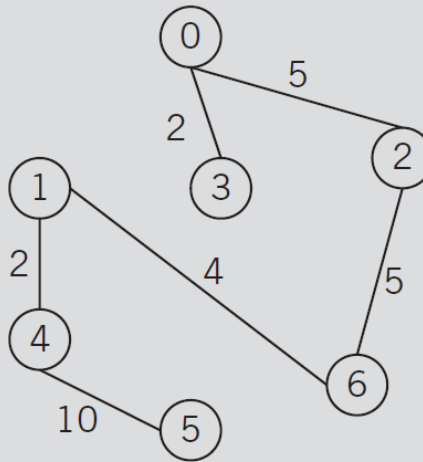


Graphs

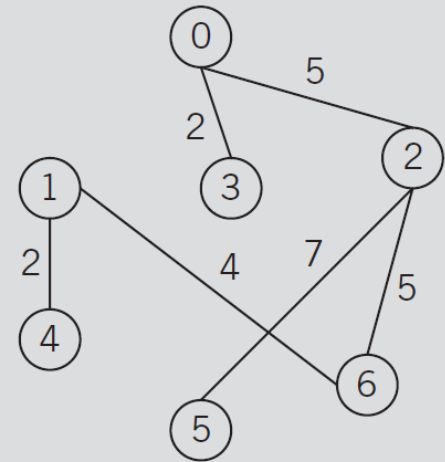
- Spanning trees from the graph on the previous slide:



(a)



(b)



(c)

Graphs

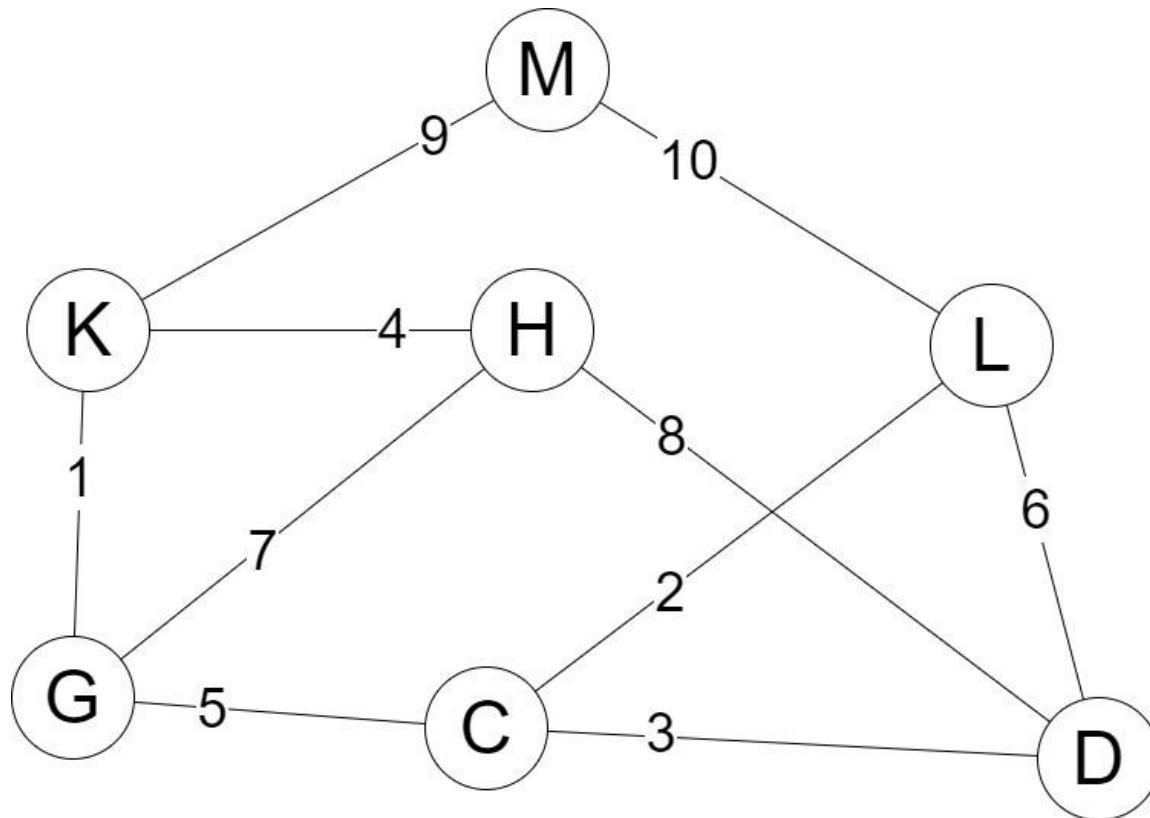
- Prim's MST algorithm:

The general form of Prim's algorithm is as follows. (Let n be the number of vertices in G .)

1. Set $V(T) = \{\text{source}\}$
2. Set $E(T) = \text{empty}$
3. for $i = 1$ to n
 - 3.1. $\text{minWeight} = \text{infinity};$
 - 3.2. for $j = 1$ to n
 - if v_j is in $V(T)$
 - for $k = 1$ to n
 - if v_k is not in T and $\text{weight}[v_j, v_k] < \text{minWeight}$
 - {
 - $\text{endVertex} = v_k;$
 - $\text{edge} = (v_j, v_k);$
 - $\text{minWeight} = \text{weight}[v_j, v_k];$
 - }
 - 3.3. $V(T) = V(T) \cup \{\text{endVertex}\};$
 - 3.4. $E(T) = E(T) \cup \{\text{edge}\};$

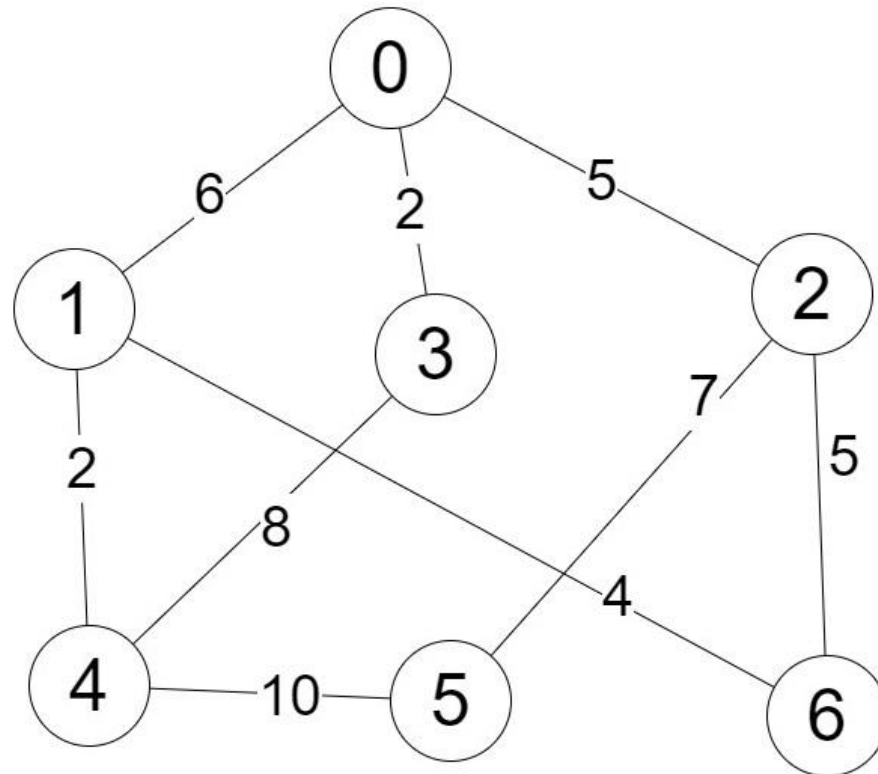
Graphs

- Prim's MST drill D1:
 - Start with node C ... identify the first 4 edges collected ...



Graphs

- Prim's MST drill D2:
 - Start with node 6 ... identify the first 4 edges collected ...



Assignment

- Read Malik chapter 11 – binary trees
- Read Malik chapter 12 - graphs